

Excellent! You want to **dual-boot Proxmox VE and VMware ESXi** on your **Dell PowerEdge R640**, using:

-  **2 × SSDs** → one for **Proxmox**, one for **ESXi**
-  **2 × HDDs** → shared storage (optional RAID 1)
-  **No virtualization overhead** — full hardware access in each OS
-  **Choose at boot time** which hypervisor to run

This is **100% possible** and a great setup for learning, testing, or lab environments.

Dual-Boot Proxmox + ESXi: Step-by-Step Guide

>  This guide assumes:

- > - You have **4 drives** (2 SSDs, 2 HDDs)
- > - You're using the **PERC RAID controller**
- > - You want **full redundancy and performance**
- > - You're okay with **reinstalling/reformatting all drives**

What You'll Need

Item Source

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| **Proxmox VE ISO** |
<https://www.proxmox.com/en/downloads> |
| **Dell Custom ESXi ISO** | [[Dell Support Site](https://www.dell.com/support)](<https://www.dell.com/support>) (search
"R640 ESXi") |
2x USB drives (or reuse one)	For bootable installers
Keyboard + Monitor or **iDRAC access**	For setup
iDRAC Enterprise (recommended)	For remote ISO mounting

🛠️ Step 1: Prepare Drives — Use Each SSD Separately

Since we want **two independent hypervisors**, we **cannot** put both SSDs in the same RAID array.

We'll use:

- **SSD 1** → **Proxmox VE**
- **SSD 2** → **ESXi**
- **HDDs** → Optional: RAID 1 for shared storage

> 🔍 Goal: Create **two separate single-drive RAID 0 arrays** (Dell's way of saying "use this drive alone")

✅ Enter PERC RAID Configuration (Ctrl+R)

1. Power on R640.
2. Press **Ctrl+R** during boot → **PERC Configuration Utility**

🛠️ Step 1.1: Create RAID 0 for SSD 1 (Proxmox)

1. Select **Controller 1**
2. Press **F2** → **Create Virtual Disk**
3. Settings:
 - **RAID Level**: `RAID 0` (Single drive)
 - **Select Physical Disk**: Choose **SSD 1**
 - **Strip Size**: `64 KB`
 - **Name**: `SSD-Proxmox`
 - **Write Policy**: `Write Back with BBU`
 - **Disk Cache**: `Enabled`

✅ Press **OK** → Confirm

🛠️ Step 1.2: Create RAID 0 for SSD 2 (ESXi)

Repeat:

1. Press **F2** → **Create Virtual Disk**
2. Settings:

- **RAID Level**: ` RAID 0`
- **Select Physical Disk**: **SSD 2**
- **Name**: ` SSD-ESXi`
- Same settings as above

 Press **OK**

🛠 Step 1.3: Create RAID 1 for HDDs (Optional – Shared Storage)

1. Press **F2** → **Create Virtual Disk**

2. Settings:

- **RAID Level**: ` RAID 1`
- **Select both HDDs**
- **Name**: ` HDD-DATA-MIRROR`
- Use ` Write Back` if BBU is healthy

 Press **OK**

Save & Exit PERC

- Press **Esc** → **Home**
- Press **F10** → **Apply Changes and Exit**
- Server reboots

🛠️ Step 2: Install Proxmox VE (First OS)

Create Bootable USB

Use **Rufus** (Windows) or `dd` to write **Proxmox ISO** to USB.

> Example `dd` command:

```
```bash
```

```
sudo dd if=proxmox-ve_*.iso of=/dev/sdX bs=1M status=progress && sync
```

```
```
```

Boot & Install Proxmox

1. Insert USB → Press **F11** → Boot from USB

2. Select **Install Proxmox VE**

3. Follow installer:

- Country, Language, Keyboard → Set
- **Target Disk**: Should show `SSD-Proxmox` (e.g., `/dev/sda`)
- ☒ **Use all space**
- Hostname: `proxmox01`
- IP, Gateway, DNS: Configure for your network
- Password: Set strong root password
- Email: `admin@lab.local`

4. Click ****Install****

Wait 5–10 mins → Reboot

> 🗑️ Remove USB after reboot

✅ Access Proxmox Web UI

- Open browser → `https://<your-ip>:8006`

- Log in with `root@pam` and your password

✅ Proxmox is now installed on ****SSD 1****

🛠️ Step 3: Install VMware ESXi (Second OS)

Now install ESXi on ****SSD 2****.

📄 Download Dell Custom ESXi ISO

Go to: https://www.dell.com/support

Search: `PowerEdge R640 ESXi`

Download: ****VMware vSphere Hypervisor (ESXi) Dell Custom Image****

>  This includes drivers for R640 (network, RAID, etc.)

Create ESXi Bootable USB

Use **Rufus** (in **DD mode**) or `dd`:

````bash`

`sudo dd if=ESXi-*.iso of=/dev/sdX bs=1M status=progress && sync`

`````

Boot & Install ESXi

1. Insert **ESXi USB**
2. Press **F11** → Boot from USB
3. Wait for installer to load

ESXi Installer Steps:

1. **Welcome** → Press **Enter** to install
2. **Select Disk**:
 - You'll see two drives:
 - `RAID 0: SSD-Proxmox` ← **DO NOT USE**
 - `RAID 0: SSD-ESXi` ←  **Select this one**
 - Highlight it → Press **Enter**

3. **Confirm overwrite** → Press **F11**

4. **Set Root Password** → Enter strong password (required)

5. **Confirm installation** → Wait 5–10 mins

6. When done → **Remove USB** → Press **Enter** to reboot

 Access ESXi Web UI

After boot:

- Watch console for IP address (or set static)
- Open browser → `https://<esxi-ip>/ui`
- Log in with `root` and password

 ESXi is now installed on **SSD 2**

 Step 4: Dual-Boot — How to Switch OS

There is **no shared bootloader** (like GRUB). Instead, use:

 Method: **F11 Boot Menu (Recommended)**

At every boot:

1. Power on server
2. Press **F11** immediately → **One-Time Boot Menu**
3. Choose:
 - **SSD-Proxmox** → Boots **Proxmox**
 - **SSD-ESXi** → Boots **ESXi**

>  You pick which hypervisor to run at startup.

 Optional: Use HDD RAID for Shared Storage

You can use the **HDD RAID 1 array** to store:

- ISOs
- VM templates
- Backups (exported VMs)

In Proxmox:

```
```bash
```

```
Format HDD RAID (if not done)
```

```
mkfs.ext4 /dev/sdb1
```

```
Mount
```

```
mkdir /mnt/pve/shared
```

```
mount /dev/sdb1 /mnt/pve/shared
```

# Add to /etc/fstab

```
echo '/dev/sdb1 /mnt/pve/shared ext4 defaults 0 0' >> /etc/fstab
```

```

Then add as **Storage** in Proxmox UI.

In ESXi:

- Go to **Storage** → **New Datastore**
- Format as **VMFS6**
- Name: `Shared-Data`
- Use for VMs, ISOs, backups

> **!** You **cannot access the same datastore from both OSes at once** — unmount in one before booting the other.

🧩 Summary: Final Layout

| Drive | Purpose | OS Access |
|-------|---------|-----------|
|-------|---------|-----------|

| | | |
|-------|-------|-------|
| ----- | ----- | ----- |
|-------|-------|-------|

| | | |
|--------------|---------------|------------------------------|
| SSD 1 | Proxmox VE OS | Only when Proxmox is running |
|--------------|---------------|------------------------------|

| | | |
|--------------|---------|---------------------------|
| SSD 2 | ESXi OS | Only when ESXi is running |
|--------------|---------|---------------------------|

| | | |
|----------------------|-------------------------------------|---------------------------|
| HDDs (RAID 1) | Shared storage (ISOs, VMs, backups) | Mount in one OS at a time |
|----------------------|-------------------------------------|---------------------------|

✅ Benefits of This Setup

- ✅ Full hardware access in both hypervisors
- ✅ No performance loss
- ✅ Safe — one OS can't break the other
- ✅ Great for **learning, testing, comparing Proxmox vs ESXi**
- ✅ Easy to maintain

❌ Limitations

- ❌ Cannot run both OSes at the same time
- ❌ Must reboot to switch
- ❌ Shared storage must be unmounted cleanly

📌 Pro Tips

1. **Label your SSDs** physically (e.g., "Proxmox", "ESXi")
2. **Enable iDRAC** → Set static IP → Access console remotely
3. **Update firmware** via **F10 at boot** (Lifecycle Controller)

4. **Backup VMs** regularly — RAID is not a backup!

Need Help With?

Let me know:

- Whether you want to **automate boot selection**
- How to **export/import VMs** between Proxmox and ESXi
- Setting up **pfSense, Windows VMs, or Kubernetes**
- Using **ZFS instead of RAID**

I'll give you exact commands and configs!  